

In many complex, dynamical information systems, *affinity* is the logical mechanism facilitating the propagation of information. Complex systems refer to those who are composed of many small and primitive elements who obey simple, (usually) deterministic rules, but who achieve global behavior that is coherent and sophisticated. Commonly, it is hard to trace a satisfying mapping from the small elements to the larger behavior simply because (1) there are so many small elements and (2) their rules are *so simple!* Dynamical systems are kinds of complex systems where the global behavior can adapt extraordinarily well to internal or external stimulus or information.

Some systems are complex in that they have many simple pieces that compose a powerful whole, but are not as dynamical or mysterious. Computers are a great example. Alan Turing's formulation of *computation* made it plainly clear that any possibly articulable or imaginable computation could be achieved with a vanishingly tiny toolkit of operations. Since we built computers and designed their theory ourselves, we have a strong grasp of the mapping from the tiny bit registers and the simple operations which create and use them and the resulting large-scale computations they achieve. In this way, computers are great idealized examples of complex systems. We can use computation to understand complex systems that we did not build.

The three most fascinating (to me) kinds of complex systems which we did not manually build are: (1) evolved biological systems, (2) artificial neural networks, and (3) financial markets. The underlying principle connecting these systems (and others) is *affinity*.

In all of these systems, the individual elements cannot possibly grasp the global whole which they compose. They have tunnel vision. The whole is simply emergent from the activity of the denizens. Without any global coordination whatsoever, it is the repeated and parallel activity that is *facilitated by affinity* that composes into a whole system. When I say affinity, I am referring to both actual identity, but also *mirror* affinity, which is to say *complementarity*.

In biology, the necessary prerequisite for interaction of two elements is chemical complementarity. The charges must match for molecules to interact. This law is the product of chemistry and thus physics; it is invariant. Thus, the complex systems of biology had to learn to utilize this mechanism as the minimal facilitator of information. Biological systems *grew out of* chemical laws of affinity. These laws were not *chosen* as facilitators of the system, they endowed the system itself.

In neural networks, the Dot Product is the kind of mathematical operation which facilitates the propagation of information. The Dot Product is a way to take two vectors and determine their directional similarity. The higher a dot-product between two vectors, the more they face in the exact same direction; a dot product of 0 means the vectors are perfectly orthogonal; a negative dot product means the vectors are pointing away from each other. A neural network is a massive information-filtering mechanism. Precisely, it is a series of cascading templates which will only activate if what comes in matches the template with a high dot product. 'Early' templates in the network make very primitive decisions with simple

templates. The results of each layer of templates (*which templates are active and which are not*) is passed as the input to the next layer. It is all facilitated by successive measures of affinity.¹

Now for neural networks, we *did* choose this operation as the facilitator. Neural networks exist in computer programs of our design.² However what we do not choose is just what each template should look for—to *what* the inputs have to complement. These templates are all learned iteratively, **such that** when they all activate on the basis of affinity with their input, the network as a whole has favorable behavior. ‘To the network’, the law of affinity is just as invariant as that chemical law of affinity that facilitates biology. The templates in the network had to learn to grow *around* the foundational principle of affinity in order to work correctly.

Biological systems grew and adapted *around* affinity by way of evolution. The DNA which prescribed all of the structures that interfere constantly on the basis of chemical affinity had to build these structures such that when they all act on the basis of affinity with their surroundings, the organism as a whole had favorable behavior. Neural networks’ templates are adjusted by the program which runs them, which uses calculus. It has a much shorter timeline than random evolution, however both systems evolve analogously. Affinity is the rule, so *use* it to do useful work. *Encode the world you want to solve* as a problem of affinity checks.

Financial markets are different in many ways from biological systems and neural networks, however they too are complex information-processing systems that are composed of imperfectly-informed and globally-ignorant denizens who can only interact if some affinity is achieved. A market is strictly a place for the exchange of goods (usually one of the goods is money). For all of the social fizz that surrounds a market, all that the market is is a set of trades. There are people, there are goods, and sometimes two people agree to trade goods at some price. The price of the buyer must match that of the seller if a trade is to be made; that is all. The ‘price’ of a stock or commodity in a market is denoted by either the most recently agreed-upon price or some kind of average between the highest offer from a buyer and the lowest demand from a seller. As whole networks, markets converge to highly efficient equilibrium prices. In well-populated markets, the margins between highest offer and lowest demand can be one single penny. Though each operation within the network is as simple as an equality of offer and demand, as a whole, these operations in parallel can consume the sharded information present in all of the people’s heads and compose them into a price. Why are markets such good information-processors? Because affinity is found between two shrewd people who both think

¹ It is worth mentioning that the dot product, an affinity measure, is used in many other advanced systems or methods of processing information. The Fourier Transform, which transforms signals from the time domain into the frequency domain, is highly important for many applications. It composes the transformation by simply quantifying the affinity of some signal with all of the possible frequencies that could be present in the signal—it computes dot products between the signal and the frequencies it wants to scan. These frequencies are templates just like those in a neural net. (Idealized, dimensionally reduced) Fourier Transforms can be achieved with one-layer neural networks. The basilar membrane in your ear essentially computes a Fourier Transform, but instead of taking in vectors of sounds and doing dot products, it has evolved to be built such that parts of the membrane will vibrate only when certain frequencies match its resonance frequency. Affinity!

² Artificial neural networks, that is. ANNs were inspired by biological neural networks and the Dot Product is an *abstraction* constructed to mimic the templatic activation of real neurons. There are no vectors coursing in your brain, just electricity and chemicals.

they are making a good trade. Markets only move when people put their money where their mouths are.

Interestingly, though, in markets, the affinity-finding itself is not what actually moves elements around. It is the *approach towards* affinity that causes markets to move. The computation and processing done by the market ends when a deal is made; it is active in the search for a deal. In biological or neural systems, affinity-finding is the fabric of change and activity; in markets, it is the force pulling at the elements which becomes the fabric of activity.

Why is affinity such a common tool for complex information-processing systems that are composed of local interactions between simple denizens? There are three key characteristics of affinity that make it such a reliable facilitator of computation:

- Locality
- Parallelizability
- Composability

Affinity is an entirely local property. It is patently concerned with the similarity of *these two elements*, not a whole system of elements at once. Whether it be via electromagnetic charge, the interaction of numbers via multiplication, or the simple agreement on a price, a measurement of affinity requires no external information other than what is being compared. Affinity almost doesn't even need to be explained or instructed. It speaks for itself.

Because so many denizens can easily compute local affinities and each computation needs no external information, affinity checks can occur in parallel. Two pairs of elements can check for affinity at once, completely separate from each other. This is vital for massively complex information-processing systems.

Finally, if you can have so many local affinity checks happening in parallel, then the products of those checks—however encoded—can be composed into larger achievements of information. This is best illustrated with the templates in a neural network. If there are independent templates in a network that scan, say, an inputted image for eyes, ears, a nose, a mouth, and hair, they can compose to determine a much larger result. Many 'higher level' discriminations or pieces of information can be defined in terms of a *conjunction* of many lower level discriminations.

Our eyes process everything we see into a uniform set of *basic* activations, but compose into every coherent *thing* you've ever seen. That composition, too, happens in the same neural jungle as that which amplifies some impulse into a massive bodily action. That neural jungle, too, is composed of small molecules which are indifferently acting at the whims of affinity.

It is affinity all the way down. Complex affinity-based systems are composed of lower-order affinity-based systems. The information-processing abilities of any complex system is a function not just of the affinity checks themselves, but how they compose and interact. Because affinity does not check for anything but the intrinsic properties of whatever elements it

is concerned with, it operates based on content, not name or index. All that matters is intrinsic affinity. It doesn't matter what the context or alias is; affinity needs no labels or names or rules.